Malware Analysis Report

UNKNOWN.EXE

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Executive Summary

A binary identified as `unknown.exe` has been analyzed and determined to be a **self-deleting data exfiltration malware**. The malware exhibits conditional execution based on its ability to connect to specific domains associated with a suspected command-and-control (C2) infrastructure.

Upon successful network communication with `update.ec12-4-109-278-3-ubuntu20-04.local`, the malware attempts to connect to a secondary domain, `cdn.altimiter.local`, where it issues HTTP GET requests containing RC4-encrypted payloads via the `post` parameter. These payloads are believed to represent exfiltrated data. If the initial domain is unreachable, the malware initiates a function titled `Houdini` that triggers its self-deletion to evade detection.

Additionally, during execution, the malware creates a plaintext file (`C:\Users\Public\passwrd.txt`) containing the string `SikoMode`, which is presumed to be the RC4 decryption key.

This behavior demonstrates the malware's purpose as a **data theft utility with anti-forensic capabilities**. It does not implement persistence, further indicating its design for stealthy, one-time execution.

Recommended Remediation Methods

- Immediately quarantine and remove `unknown.exe` and `passwrd.txt` from all affected endpoints.
- Block and monitor the following domains at perimeter and endpoint network controls:
  - `http://update.ec12-4-109-278-3-ubuntu20-04.local`
  - `http://cdn.altimiter.local`
- Implement deep packet inspection (DPI) or proxy filters to detect unusual HTTP GET requests with `feed?post=` patterns.
- Develop and deploy endpoint detection rules to flag self-deleting binaries or RC4-like obfuscation patterns.
- Monitor for write attempts to `C:\Users\Public` and unexpected process activity tied to network exfiltration.

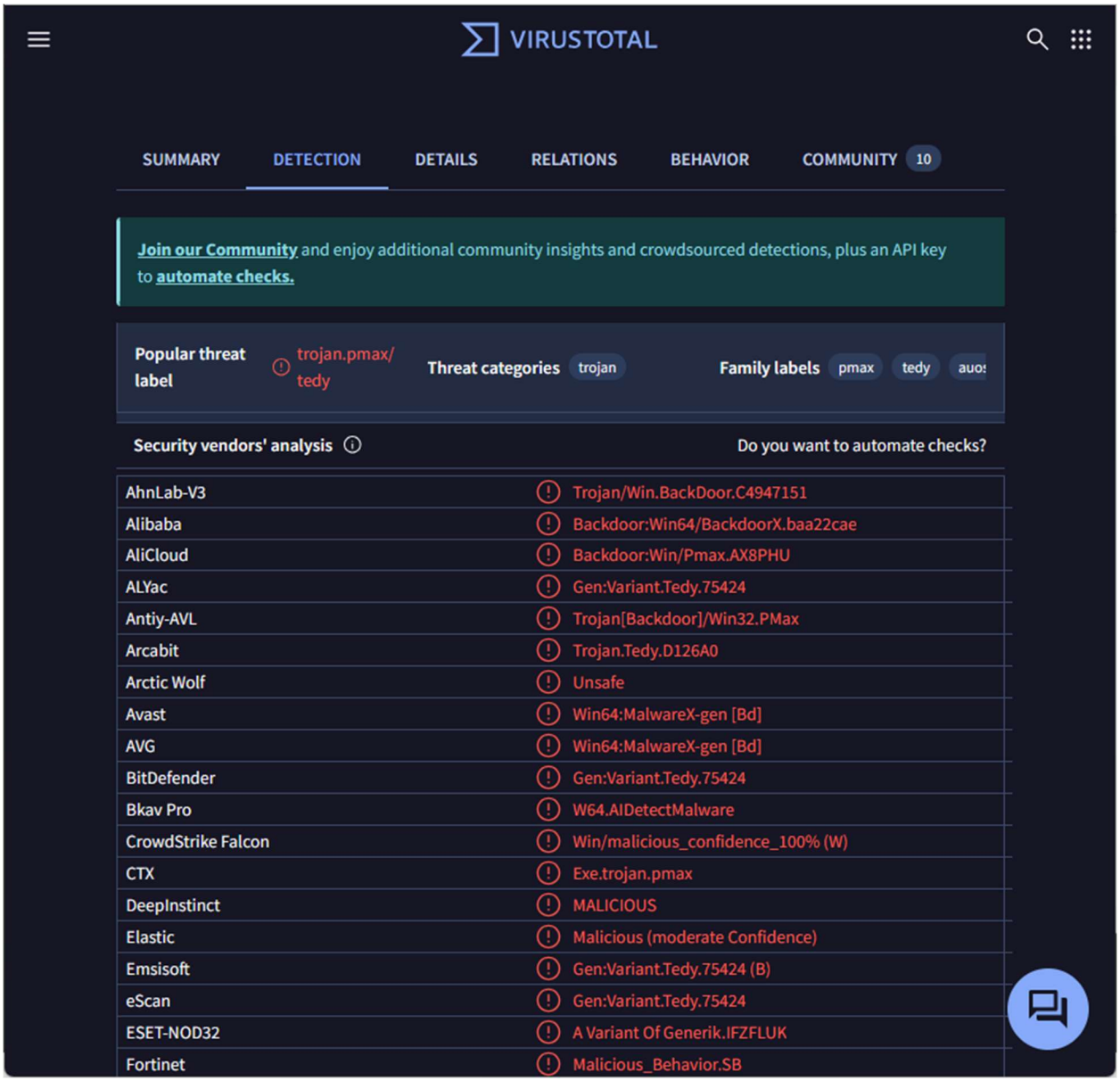
Static Analysis

Hashes

MD5	b9497ffb7e9c6f49823b95851ec874e3
SHA1	6c8f50040545d8cd9af4b51564de654266e592e3
SHA256	3aca2a08cf296f1845d6171958ef0ffd1c8bdfc3e48bdd34a605cb1f7468213e

VirusTotal

VirusTotal recognized the file hash as a hash belonging to a known backdoor/trojan:



Data Size

Virtual Size	100,376
Raw Data Size	100,864
Assumption: Malware is unpacked	

Strings

@:houdini

@Authorization

@Host

@httpclient.nim(1144, 15) `false`

@Transfer-Encoding

@Content-Type

@Content-Length

@httpclient.nim(1082, 13) `not url.contains({'\r', '\n'})` url shouldn't contain any newline characters

@http://cdn.altimiter.local/feed?post=

@Nim httpclient/1.6.2

@Desktop\cosmo.jpeg

@SikoMode

@iterators.nim(240, 11) `len(a) == L` the length of the seq changed while iterating over it

@ccc

@Mozilla/5.0

@C:\Users\Public\passwrd.txt

toRC4__OOZOOZOOZOOZOOZOnimbleZpkgsZ8267524548O49O48Z826752_51

@m..@s..@s..@s..@s..@s.nimble@spkgs@sRC4-0.1.0@sRC4.nim.c

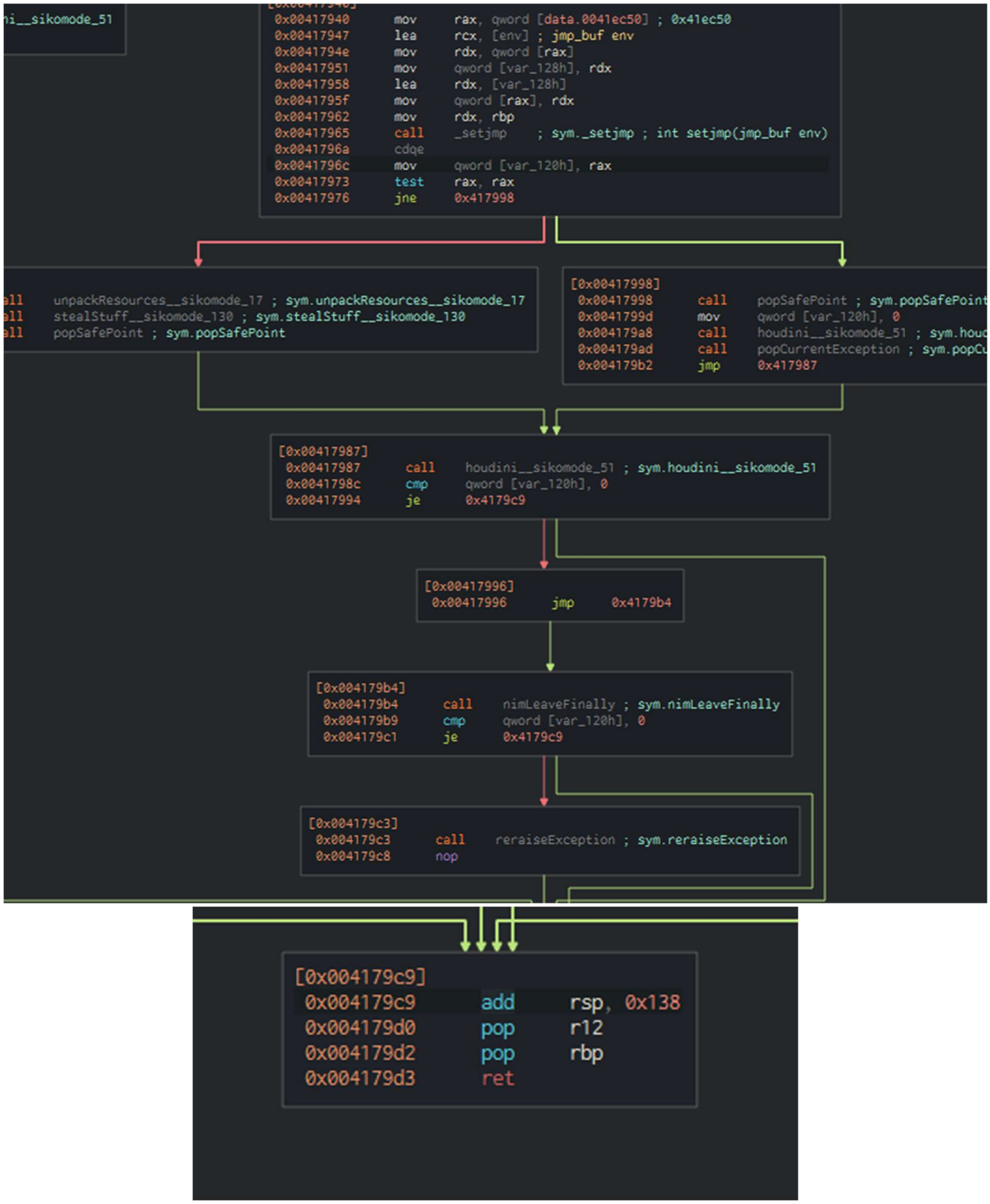
Decompiling/Disassembling

When decompiling the malware, the analyst discovered a method that appears to RC4 encrypt the data being exfiltrated:

```
[0x00409ab2]
 toRC4__OOZOOZOOZOOZOOZOnimbleZpkgsZ826752454804904 8Z826752_51(int64_t arg1, int64_t arg2);
 ; arg int64_t arg1 @ rcx
 ; arg int64_t arg2 @ rdx
 ; var int64_t var_888h @ stack - 0x888
 ; var int64_t var_880h @ stack - 0x880
 ; var int64_t var_878h @ stack - 0x878
 ; var int64_t var_870h @ stack - 0x870
 ; var int64_t var_868h @ stack - 0x868
 ; var int64_t var_860h @ stack - 0x860
 ; var int64_t var_858h @ stack - 0x858
 ; var int64_t var_850h @ stack - 0x850
 ; var int64_t var_848h @ stack - 0x848
 0x00409ab2      push    r15
 0x00409ab4      push    r14
 0x00409ab6      push    r13
 0x00409ab8      push    r12
 0x00409aba      push    rbp
 0x00409abb      push    rdi
 0x00409abc      push    rsi
 0x00409abd      push    rbx
 0x00409abe      sub     rsp, 0x868
 0x00409ac5      xor     eax, eax
 0x00409ac7      xor     r12d, r12d
 0x00409aca      mov     rbx, rdx    ; arg2
 0x00409acd      lea     rdx, [var_848h] ; int64_t arg2
 0x00409ad2      mov     r8, rcx     ; arg1
 0x00409ad5      mov     ecx, 0x200 ; 512
 0x00409ada      mov     rdi, rdx
 0x00409add      rep     stosd dword [rdi], eax
 0x00409adf      mov     rcx, r8     ; int64_t arg1
 0x00409ae2      call    genKeystream__OOZOOZOOZOOZOOZOnimbleZpkgsZ826752454804904 8Z826752_2 ; sym...
 0x00409ae7      test    rbx, rbx
 0x00409aea      je      0x409cd0
```

The analyst discovered a method called *Houdini*. This method exits the binary and deletes itself from disk:

```
[0x00417936]
 0x00417936      call    houdini__sikomode_51 ; sym.houdini__sikomode_51
 0x0041793b      jmp     0x4179c9
```


0x00417940 mov rax, qword [data.0041ec50] ; 0x41ec50
0x00417947 lea rcx, [env] ; jmp_buf env
0x0041794e mov rdx, qword [rax]
0x00417951 mov qword [var_128h], rdx
0x00417958 lea rdx, [var_128h]
0x0041795f mov qword [rax], rdx
0x00417962 mov rdx, rbp
0x00417965 call _setjmp ; sym._setjmp ; int setjmp(jmp_buf env)
0x0041796a cdqe
0x0041796c mov qword [var_120h], rax
0x00417973 test rax, rax
0x00417976 jne 0x417998
unpackResources__sikomode_17 ; sym.unpackResources__sikomode_17
stealStuff__sikomode_130 ; sym.stealStuff__sikomode_130
popSafePoint ; sym.popSafePoint
[0x00417998]
0x00417998 call popSafePoint ; sym.popSafePoint
0x0041799d mov qword [var_120h], 0
0x004179a8 call houdini__sikomode_51 ; sym.houd
0x004179ad call popCurrentException ; sym.popCu
0x004179b2 jmp 0x417987
[0x00417987]
0x00417987 call houdini__sikomode_51 ; sym.houdini__sikomode_51
0x0041798c cmp qword [var_120h], 0
0x00417994 je 0x4179c9
[0x00417996]
0x00417996 jmp 0x4179b4
[0x004179b4]
0x004179b4 call nimLeaveFinally ; sym.nimLeaveFinally
0x004179b9 cmp qword [var_120h], 0
0x004179c1 je 0x4179c9
[0x004179c3]
0x004179c3 call reraiseException ; sym.reraiseException
0x004179c8 nop
[0x004179c9]
0x004179c9 add rsp, 0x138
0x004179d0 pop r12
0x004179d2 pop rbp
0x004179d3 ret

Dynamic Analysis

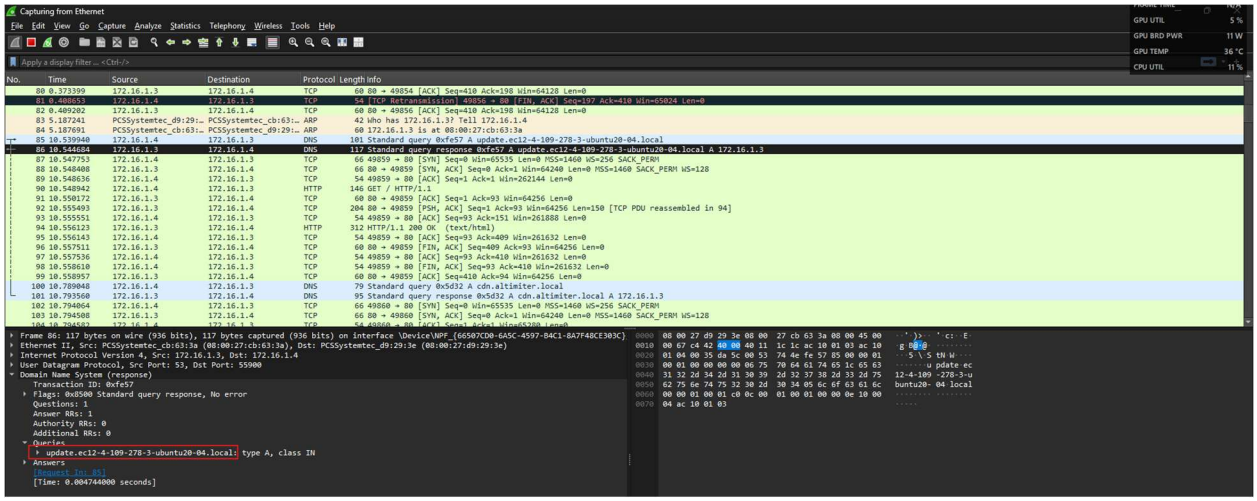
Initial Detonation

During the initial detonation there was not any immediate sign of activity

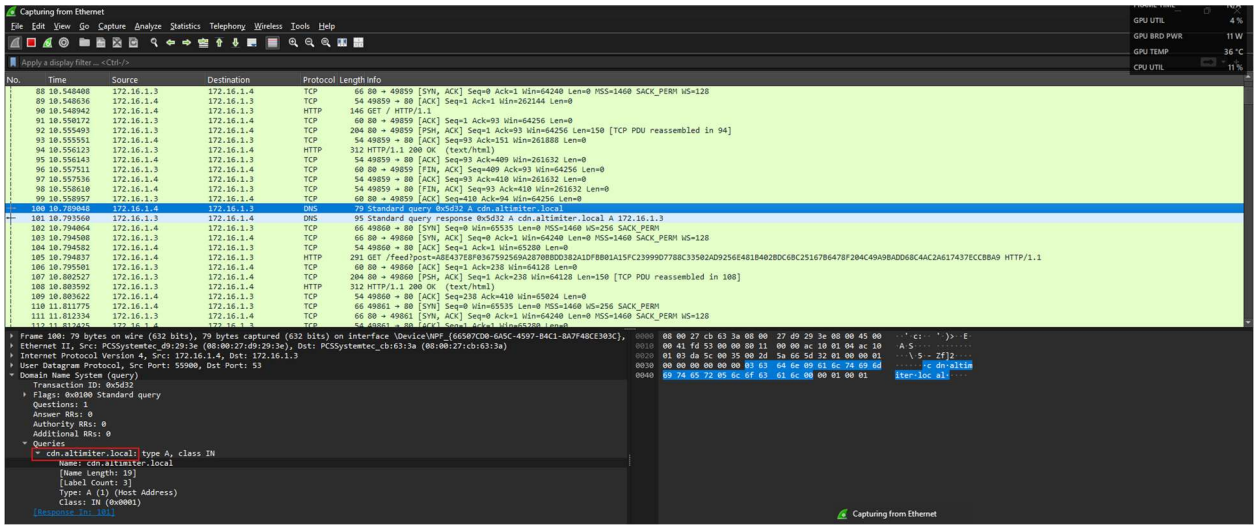
Network-Based Signatures

WireShark

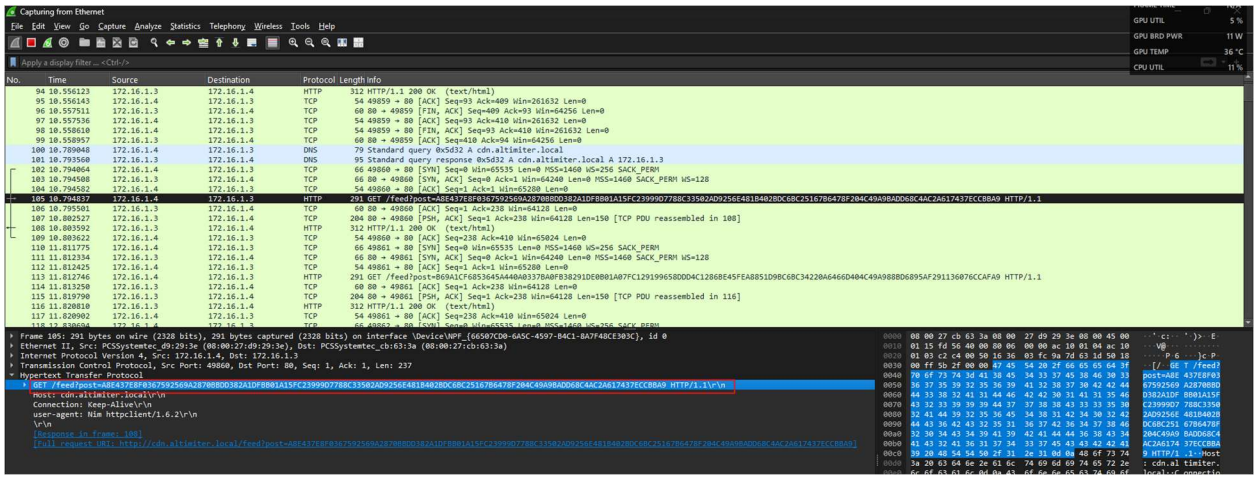
Within WireShark we can see a DNS query for an EC2 instance *update.ec12-4-109-278-3-ubuntu20-04.local*:



The analyst was able to identify traffic between the local host and the remote EC2 instance. This is immediately followed by another DNS query for *cdn.altimiter.local* domain:

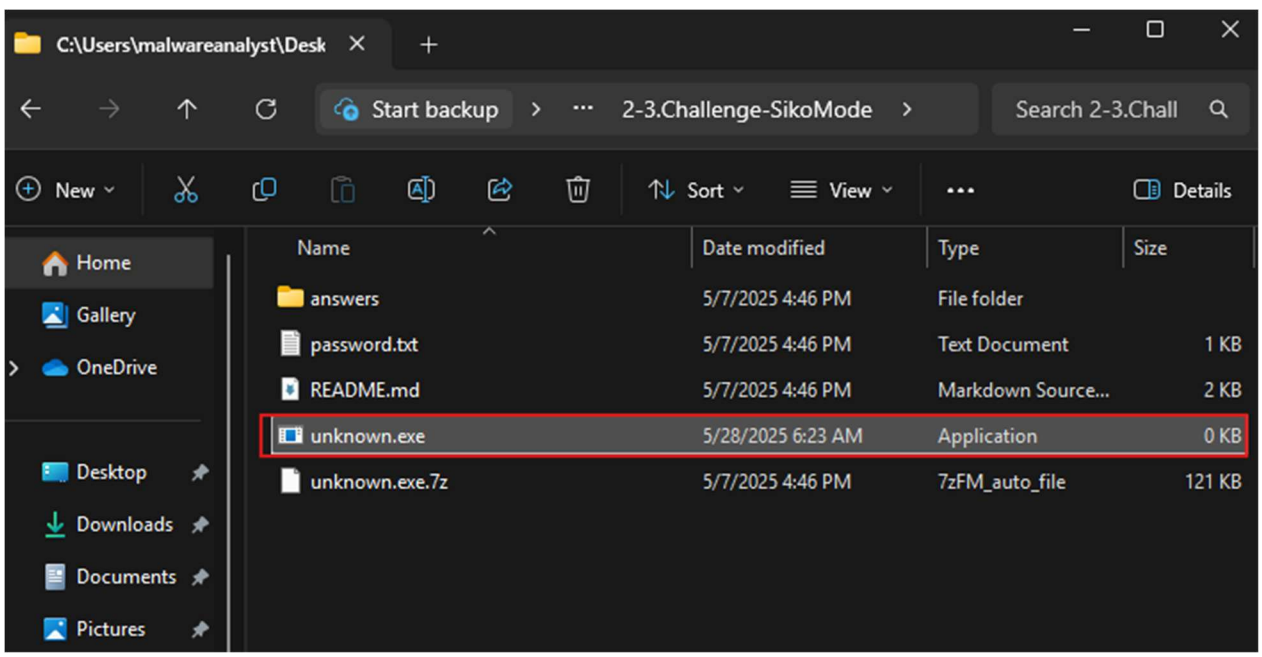


The analyst also discovered another HTTP GET request to the /feed subdirectory with a post parameter and string of characters for the value:



Host-Based Signatures

When the malware runs and it cannot make an HTTP connection the malware self-deletes itself:



ProcMon

Within ProcMon the analyst discovered the creation of the password.txt file:

Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

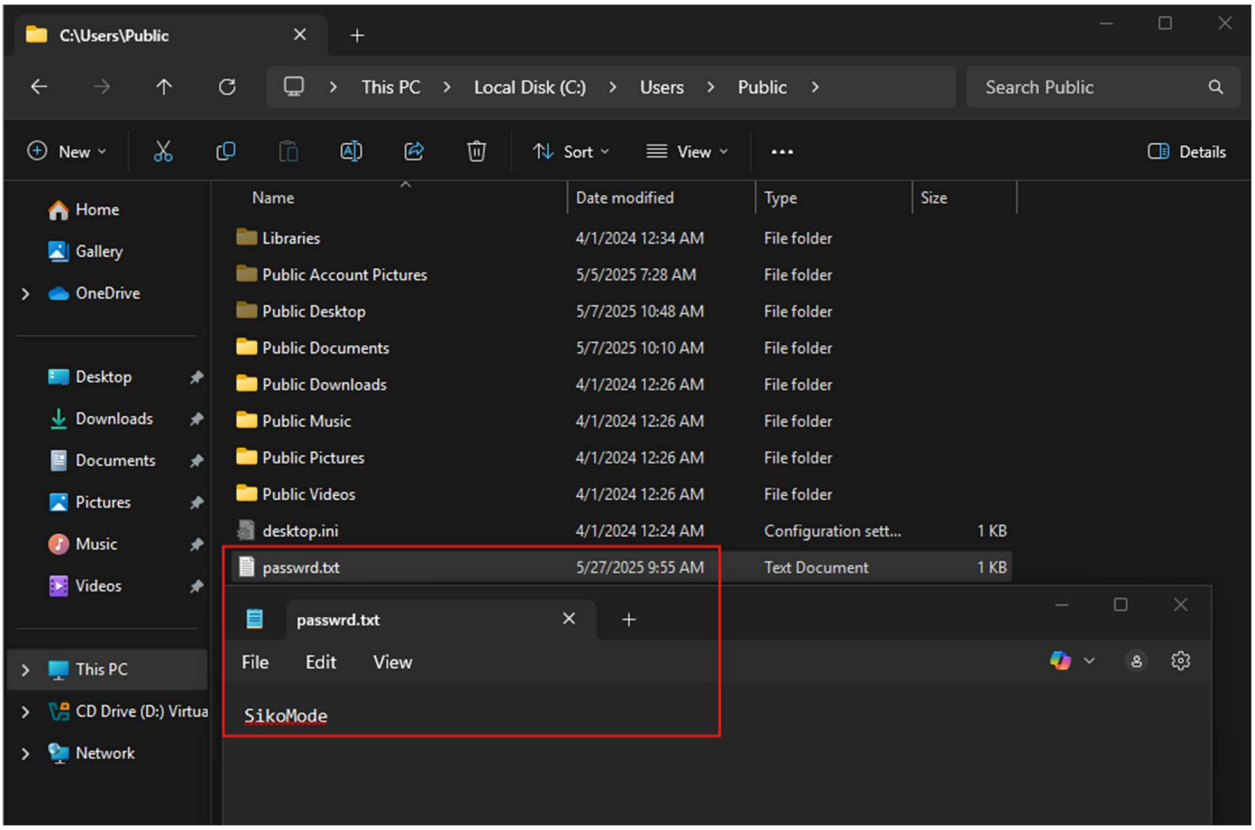
Time ...	Process Name	PID	Operation	Path	Result	Detail
6:33:2...	unknown.exe	4920	CreateFile	C:\Users\Public\passwrd.txt	SUCCESS	Desired Access: G...
6:33:2...	unknown.exe	4920	WriteFile	C:\Users\Public\passwrd.txt	SUCCESS	Offset: 0, Length: 8...
6:33:2...	unknown.exe	4920	CloseFile	C:\Users\Public\passwrd.txt	SUCCESS	
6:35:5...	unknown.exe	6264	CreateFile	C:\Users\Public\passwrd.txt	SUCCESS	Desired Access: G...
6:35:5...	unknown.exe	6264	WriteFile	C:\Users\Public\passwrd.txt	SUCCESS	Offset: 0, Length: 8...
6:35:5...	unknown.exe	6264	CloseFile	C:\Users\Public\passwrd.txt	SUCCESS	
6:38:0...	unknown.exe	4996	CreateFile	C:\Users\Public\passwrd.txt	SUCCESS	Desired Access: G...
6:38:0...	unknown.exe	4996	WriteFile	C:\Users\Public\passwrd.txt	SUCCESS	Offset: 0, Length: 8...
6:38:0...	unknown.exe	4996	CloseFile	C:\Users\Public\passwrd.txt	SUCCESS	
7:08:1...	unknown.exe	3496	CreateFile	C:\Users\Public\passwrd.txt	SUCCESS	Desired Access: G...
7:08:1...	unknown.exe	3496	WriteFile	C:\Users\Public\passwrd.txt	SUCCESS	Offset: 0, Length: 8...
7:08:1...	unknown.exe	3496	CloseFile	C:\Users\Public\passwrd.txt	SUCCESS	
7:12:4...	unknown.exe	9492	CreateFile	C:\Users\Public\passwrd.txt	SUCCESS	Desired Access: G...
7:12:4...	unknown.exe	9492	WriteFile	C:\Users\Public\passwrd.txt	SUCCESS	Offset: 0, Length: 8...
7:12:4...	unknown.exe	9492	CloseFile	C:\Users\Public\passwrd.txt	SUCCESS	

TCPView

Within TCPView the analyst discovered several TCP connections over port 80 to REMnux machine:

Process Name	Process ID	Protocol	State	Local Address	Local Port	Remo
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52311	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52312	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52313	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52314	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52315	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52316	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52317	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52318	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52319	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52320	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52321	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52322	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52323	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52324	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52325	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52326	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52327	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52328	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52329	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52330	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52331	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52332	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52333	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52334	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52335	172.16
unknown.exe	4888	TCP	Close Wait	172.16.1.4	52336	172.16

Correlation

Based on the network signatures and the host-based signatures we can see that several TCP connections are made over port 80 around the time the malware is exfiltrating data to the remote host.

Combining the information gathered from Static Analysis, with running the malware we can see that a passwrd.txt file was created in the C:\Users\Public directory: